

Problem Set 2

Prof. Oke

CEE 260/MIE 273: Probability & Statistics in Civil Engineering

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*Due Tuesday, September 22, 2026 at 12:59 PM as PDF uploaded via Gradescope. If it helps and if possible, you can write your responses directly on this document and upload it instead. **Show as much work as possible in order to get FULL credit.** There *FOUR* problems*

Problem 1 (5 points)

Respond “T” (*True*) or “F” (*False*) to the following statements. Use the boxes provided. Each response is worth 1 point.

- (i) ☐ An impossible event has a probability less than or equal to 0.
- (ii) ☐ A certain event is equivalent to the sample space of outcomes.
- (iii) ☐ If A and B are mutually exclusive, then $P(AB) \neq 0$.
- (iv) ☐ Given an event E , then the probability of its complement can be given as $P(\overline{E}) = 1 + P(E)$
- (v) ☐ A and B are collectively exhaustive events. If $P(B) = 0.4$, then $P(A) = 0.6$.

Problem 2 (7 points)

Show your work/explanations in order to receive full credit. Single number answers are not sufficient.

- [2] (a) License plates in a certain state consist of 3 letters followed by 3 digits. How many different license plates can be manufactured?
- [2] (b) Your computer password must have only 6 characters. Each character can be a lowercase letter (e.g. a), an uppercase letter (e.g. A) or a digit (e.g. 0). How many valid passwords could you possibly create?
- [3] (c) Your operating system is tightening its security protocols and asks you to update your password. It now requires that in addition to the earlier rules, you must now have at least one uppercase letter in your computer password. How many valid passwords could you possibly now create?

Problem 3 (8 points)

Data collected at elementary schools in DeKalb County, GA suggest that each year roughly 25% of students miss exactly one day of school, 15% miss 2 days, and 30% miss 3 or more days due to sickness.

Example

Let X be the number of school days missed in a year.

The probability that a student chosen at random misses 2 or more days of school in a year is given by:

$$P(X \geq 2) = P(X = 2) + P(X \geq 3) = 0.15 + 0.30 = 0.45$$

- (a) What is the probability that a student chosen at random does not miss any days of school due to sickness this year?
- (b) What is the probability that a student chosen at random misses no more than one day?
- (c) What is the probability that a student chosen at random misses at least one day?
- (d) What is the probability that a student chosen at random misses one or two days of school a year?

Problem 4 (*MATLAB/Python*)

(Python version to be provided before midnight on Thursday)

Objectives

- Compute permutations and combinations
- Learn how to import datasets
- Compute probabilities
- Create descriptive plots of datasets
- Compute summary statistics

Requirements

You must have a working installation of MATLAB, preferably the most recent version. Please edit and run your `.m` file in the same directory as the supplied data (which you will download from Moodle). This way, you will only need to use the filenames when loading data into MATLAB.

Submission instructions

Use the template provided to write your code. Feel free to modify to your liking. Append your name (`LASTNAME_FIRSTNAME_`) to `HW1.m` with an underscore (deleting `_template` from the filename) and then submit. Note that MATLAB filenames must begin with a letter and contain only letters, numbers and underscores. Any data or resources provided for an assignment will be available to the graders so you only need to upload an `.m` file.

Exercise 1.1 Summary statistics and probabilities

The files `2013Jan.Baltimore.MeanTempF.csv` and `2014Jan.Baltimore.MeanTempF.csv` (contained in the archive `HW1data.zip`) contain average daily temperatures (in Fahrenheit) recorded in Baltimore during the months January 2013 and January 2014, respectively.

- (a) Use a built-in MATLAB function (`csvread` is a good one) to read the January 2013 values into an array `X` and the January 2014 values into an array `Y`. (The first task has been completed in the template.)
- [1] (b) Replace `??` in line 20 with the appropriate value to find the number of days in January 2013 with an average temperature of 30F.
- [1] (c) Now uncomment the other lines to find the probability that the mean temperature in January 2013 was 30F and display your answer using the `fprintf` statement.
- [1] (d) Write a one line statement in your script to compute the probability that the mean temperature in January 2013 was *not* 30F.
- [1] (e) Display your answer to part (d) using `fprintf`.
- [3] (f) Now compute the probability that the mean daily temperature in January 2014 was greater than 32F but *no more* than 35F. Hint: to find the number of occurrences required, use:

```
nnz(Y > 32 & Y <= 35)
```

Display your answer using `fprintf`.

(g) Create the following plots:

- (i) Histogram of X [2]
- (ii) Histogram of Y [2]
- (iii) Scatterplot of Y versus X . [2]

Make sure that in each plot, the x -axis and the y -axis are both labeled. Also be sure to include titles for each plot. When you or the TA runs your script, all three plots must be displayed simultaneously. Points will be deducted in each case where these instructions are not followed. Consult `example2.m` from Module 1 for examples of creating scatterplots and labeling axes and plots. `example1.m` shows how to plot histograms.

- (h) Use MATLAB to find the mean and standard deviation of the temperatures in both X and Y . [4]
Display your answers using two `fprintf` statements. (The first one has been commented out for you. Note the usage of square brackets when breaking up a long quote in MATLAB over multiple lines.)